

QBITEL BRIDGE

Critical Infrastructure & ICS/SCADA

Deployment Checklist

10 Phases | 120-Day Deployment | Zero Downtime

Safety-First | NERC CIP | IEC 62443 | IEC 61508 Compatible

10

Phases

120

Days

0

Downtime

IEC 61508

Safe

DEPLOYMENT OVERVIEW

This checklist guides the deployment of QBITEL Bridge across critical infrastructure environments. It is structured in 10 phases spanning 120 days, designed for zero operational downtime. Each phase requires explicit sign-off from designated roles before proceeding. Safety system boundaries (IEC 61508 SIL 3/4) are formally documented in Phase 4 and require Safety Officer written approval before any active protection is enabled.

Phase	Name	Duration	Key Milestone
0	Pre-Engagement & Safety Officer Alignment	Days 1-5	Safety boundaries documented
1	Passive OT Protocol Discovery	Days 6-14	Asset inventory complete
2	Zone & Conduit Mapping (IEC 62443)	Days 15-22	Zone model approved
3	Infrastructure Readiness	Days 23-32	HSM and Ollama deployed
4	Safety System Identification & Exclusion	Days 33-38	Safety Officer sign-off
5	PLC Command Authentication Rollout	Days 39-55	Auth active on all PLCs
6	Protocol Protection Activation	Days 56-68	All protocols protected
7	NERC CIP Control Mapping	Days 69-80	Compliance mapped
8	Monitoring & Alerting Configuration	Days 81-90	SOC dashboard live
9	Compliance Validation	Days 91-105	Evidence package approved
10	Operational Handover	Days 106-120	Go-live sign-off

PHASE

0

Pre-Engagement & Safety Officer Alignment

Days 1-5

Owner: Project Manager / CISO

☐ Identify Asset Owner, Operations Manager, Safety Officer, and NERC CIP Compliance Lead

CRITICAL

☐ Complete OT asset inventory (preliminary) from existing documentation

- Review engineering drawings, P&ID diagrams, network diagrams
- Identify known PLCs, RTUs, HMIs, historians, SCADA servers

☐ Review existing NERC CIP AOP/CIP documentation and compliance status

- Identify current CIP-002 BES Cyber System classification
- Review last NERC CIP audit findings and remediation status

☐ Identify safety system boundaries (SIL 3/4 exclusion zones)

CRITICAL

- Document all IEC 61508 certified SIS systems by tag number
- Identify SIS network segments for passive-only designation

- ☐ Map IT/OT network topology (logical and physical)
 - Document existing firewall rules and DMZ architecture
 - Identify all remote access paths into OT network

- ☐ Schedule change windows for active protection phases (Phases 5, 6)
 - Confirm maintenance window schedule for next 90 days
 - Document change freeze periods that must be avoided

IMPORTANT

- ☐ Obtain project charter approval and stakeholder sign-off

CRITICAL

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

Asset Owner: _____ **Date:** _____
Signature

Operations Manager: _____ **Date:** _____
Signature

Safety Officer: _____ **Date:** _____
Signature

NERC CIP Compliance Lead: _____ **Date:** _____
Signature

PHASE

1

Passive OT Protocol Discovery

Days 6-14

Owner: OT Security Engineer

- ☐ Deploy passive network tap on target OT segment(s) - read-only, no traffic injection
 - Verify tap is receive-only by design (no transmit capability)
 - Confirm no impact on existing network traffic before proceeding

CRITICAL

- ☐ Run QBITEL discovery on target OT segments
 - Modbus TCP/RTU discovery and device enumeration
 - DNP3 master/outstation mapping
 - IEC 61850 GOOSE/SV identification
 - OPC UA server/client mapping

- ☐ Generate asset inventory report
 - Every PLC, RTU, HMI, historian, engineering workstation
 - Protocol version, firmware (where detectable), communication partners

- ☐ Identify unencrypted and unauthenticated protocol flows
 - Flag all Modbus without authentication
 - Flag all DNP3 without SAV5
 - Flag IEC 61850 GOOSE without HMAC

IMPORTANT

- ☐ Map BES Cyber Systems per NERC CIP CIP-002
 - Draft BES Cyber System classification for each discovered asset
 - Submit to NERC CIP Compliance Lead for review

- ☐ **Validate no impact on SCADA polling rates**
 - Monitor SCADA polling latency before and during tap deployment
 - Confirm no polling timeouts or communication errors

CRITICAL

- ☐ Produce vulnerability assessment and risk prioritization

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

OT Security Engineer: _____ Date: _____
Signature

Operations Manager: _____ Date: _____
Signature

NERC CIP Compliance Lead: _____ Date: _____
Signature

PHASE**2****Zone & Conduit Mapping (IEC 62443)**

Days 15-22

Owner: OT Security Architect

- ☐ **Classify all assets into IEC 62443 security zones**
 - Safety Zone: SIS systems (SIL 3/4) - passive only
 - Control Zone: PLCs, RTUs, field controllers
 - Supervisory Zone: SCADA servers, HMIs, historians
 - Enterprise Zone: business network, vendor access

CRITICAL

- ☐ **Define conduit security requirements per zone boundary**
 - Control Zone <-> Supervisory Zone: authenticated, encrypted
 - Supervisory Zone <-> Enterprise: DMZ, strictly filtered
 - Any zone -> Safety Zone: passive monitoring only, no active enforcement

- ☐ **Document trust boundaries for NERC CIP ESP mapping**
 - Map IEC 62443 zones to NERC CIP ESP boundaries
 - Identify all ESP ingress/egress points

- ☐ **Identify high-risk conduits requiring priority protection**
 - Internet-facing vendor remote access connections
 - Historian data links to corporate network
 - Engineering workstation connections to control zone

IMPORTANT

- ☐ Design DMZ architecture for IT/OT boundary
 - Define allowed data flows through DMZ
 - Identify historian replication and SCADA-to-corporate data flows

- ☐ Produce Zone and Conduit design document for approval

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

OT Security Architect: _____ Date: _____
Signature

Operations Manager: _____ Date: _____
Signature

CISO: _____ Date: _____
Signature

PHASE**3****Infrastructure Readiness**

Days 23-32

Owner: OT Infrastructure Engineer

- ☐ **Provision on-premise HSM (Thales Luna Network HSM or equivalent, FIPS 140-3 Level 3)**

CRITICAL

- Complete HSM installation and initialization
- Conduct air-gapped key ceremony with dual-person integrity
- Test HSM connectivity from QBITEL management server

- ☐ **Deploy air-gapped Ollama LLM instance on management server**

- Install Ollama and download required models to local storage
- Verify no external API calls (monitor network traffic)
- Test threat analysis queries against local model

- ☐ **Configure historian server integration**
 - OSIsoft PI System: configure AF server connection and tag subscription
 - GE Proficy or Honeywell PHD: configure data feed for physics model
 - Validate process data is flowing to QBITEL physics engine

- ☐ **Set up OT jump server with MFA for privileged access**
 - Deploy dedicated jump server in OT DMZ
 - Configure MFA (hardware token preferred for OT environments)
 - Document access procedure and emergency bypass

IMPORTANT

- ☐ **Verify air-gap integrity on OT network**
 - Confirm no internet connectivity on OT network segments
 - Verify no unauthorized wireless access points
 - Document all authorized remote access paths with approval

CRITICAL

- ☐ Validate management server meets QBITEL performance requirements

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

OT Infrastructure Engineer: _____ Date: _____
Signature

CISO: _____ Date: _____
Signature

Operations Manager: _____ Date: _____
Signature

PHASE

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Safety System Identification & Exclusion

Days 33-38

Owner: Safety Officer (REQUIRED)

- ☐ **Formally document SIS boundary (IEC 61508 SIL 3/4 certified systems)** CRITICAL
- List all SIS systems by tag number, vendor, SIL rating, and network segment
 - Include: Emergency Shutdown Systems, Fire and Gas Systems, High Integrity Pressure Protection
 - Cross-reference with process safety documentation (HAZOP, SIL assessment)

- ☐ **Configure QBITEL Safety Exclusion Zones (passive-only for all SIS segments)** CRITICAL
- Configure each SIS network segment as Exclusion Zone in QBITEL policy engine
 - Verify no active protection rules apply to any SIS network segment
 - Test that no QBITEL traffic is generated toward SIS network segments

- ☐ **Test that no QBITEL active protection affects SIS timing** CRITICAL
- Monitor SIS communication timing before and after QBITEL tap deployment
 - Confirm no change in SIS polling latency or communication patterns
 - Document timing validation results for Safety Officer review

- ☐ **Configure SIS anomaly escalation to human operators only**
- Any SIS anomaly: escalate to Safety Officer and Operations Manager
 - Confirm no automated response actions are configured for SIS segments
 - Test escalation path with simulated anomaly event

- ☐ **Review QBITEL configuration against IEC 61508 requirements**
- Confirm QBITEL does not modify SIS firmware or configuration
 - Confirm QBITEL operates outside the SIS safety certification boundary
 - Document configuration for Safety Officer review

- ☐ **Obtain Safety Officer written approval for SIS boundary configuration** CRITICAL
- Safety Officer reviews and signs SIS Exclusion Zone configuration document
 - Document approval in project record
 - MANDATORY GATE: Active protection phases cannot proceed without this sign-off

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

Safety Officer (MANDATORY): _____

Date: _____

Signature

Operations Manager: _____

Date: _____

Signature

OT Security Engineer: _____

Date: _____

Signature

PHASE

5

PLC Command Authentication Rollout

Days 39-55

Owner: OT Security Engineer



Deploy ML-DSA-65 signing on Modbus masters (non-SIS first, per change window)

IMPORTANT

- Start with lowest-risk, non-critical OT segment
- Follow change management procedure for each segment activation
- Maintain manual rollback capability throughout rollout



Test command authentication on isolated test PLC before production rollout

CRITICAL

- Validate ML-DSA-65 signed command accepted by verification layer
- Validate unsigned command rejected and SOC alert generated
- Validate replayed command rejected



Validate <1ms overhead on production control loops

CRITICAL

- Measure SCADA polling latency before activation
- Measure SCADA polling latency after activation
- Confirm <1ms overhead and <100us jitter on all measured loops



Roll out command authentication to remaining PLCs/RTUs by zone

- Control Zone PLCs: activate per zone following change window schedule
- RTUs (DNP3): activate on each master/outstation pair
- Engineering workstations: enforce certificate-based authentication



Test replay attack prevention (timestamp/nonce mechanism)

- Capture a valid signed command
- Replay the captured command after the nonce window
- Confirm replay is rejected and SOC alert generated



Validate physics-aware anomaly detection baseline established

- Confirm historian data is feeding physics model
- Verify at least 72 hours of baseline data collected
- Test that a simulated physics violation generates an alert



Produce rollout completion report documenting all activated segments

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

OT Security Engineer: _____

Date: _____

Signature

Operations Manager: _____

Date: _____

Signature

NERC CIP Compliance Lead: _____

Date: _____

Signature

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Protocol Protection Activation

Days 56-68

Owner: OT Security Engineer



Enable IEC 61850 GOOSE/SV authentication (IEC 62351)

CRITICAL

- Configure ML-DSA-65 GOOSE authentication per substation/segment
- Validate <100us overhead on GOOSE messages (well within 4ms protection requirement)
- Test protection relay operation with authenticated GOOSE traffic



Enable DNP3 Secure Authentication v5 wrapping where native SAV5 not available

- Configure QBITEL DNP3 SAV5 proxy on master stations
- Validate all outstation communication continues normally
- Test unauthorized DNP3 command rejection



Enable OPC UA security profile (SignAndEncrypt mode)

- Configure QBITEL OPC UA security proxy for legacy clients
- Validate all OPC UA client connections continue to function
- Confirm OPC UA data flows are encrypted and authenticated



Configure BACnet/IP and EtherNet/IP protection

- Enable BACnet anomaly detection and command filtering
- Enable EtherNet/IP CIP messaging authentication where feasible
- Validate Rockwell/Allen-Bradley PLC communication unaffected



Validate all protocol changes against SCADA polling requirements

CRITICAL

- Run 24-hour soak test on each protocol after activation
- Monitor for any polling timeouts, communication errors, or latency increases
- Confirm SCADA operator reports no issues



Document protocol protection configuration baseline

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

OT Security Engineer: _____ **Date:** _____
Signature

SCADA Engineer: _____ **Date:** _____
Signature

Operations Manager: _____ **Date:** _____
Signature

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NERC CIP Control Mapping

Days 69-80

Owner: NERC CIP Compliance Lead

- ☐ Map QBITEL capabilities to CIP-005 (Electronic Security Perimeter)
- Document ESP boundary monitoring coverage
 - Map zone/conduit configuration to CIP-005 requirements
 - Verify all ESP ingress/egress points are monitored

- ☐ Map to CIP-007 (System Security Management - ports, patches, access)
- Document port and service monitoring coverage
 - Map security event logging to CIP-007 requirements
 - Verify failed authentication events are captured and alerted

- ☐ Map to CIP-010 (Change Management - baseline monitoring)
- Configure CIP-010 baseline deviation detection
 - Test that new unauthorized device triggers CIP-010 alert
 - Document configuration baseline and deviation thresholds

- ☐ Map to CIP-011 (Information Protection - encryption)
- Document PQC encryption coverage for BES Cyber System communications
 - Map historian data protection to CIP-011 requirements
 - Document any unprotected flows with risk justification

- ☐ Configure automated CIP evidence collection for all mapped requirements
- Verify telemetry is being collected for each CIP requirement
 - Test evidence package generation for spot-check of data quality
 - Configure evidence retention period per NERC CIP requirements (6 years)

IMPORTANT

- ☐ Test NERC CIP report generation (<10 minutes)
- Trigger report generation and measure time to completion
 - Review report structure and evidence quality with Compliance Lead
 - Confirm report format meets auditor expectations

- ☐ Produce NERC CIP control mapping document for audit file

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

NERC CIP Compliance Lead: _____ **Date:** _____
Signature

CISO: _____ **Date:** _____
Signature

External NERC CIP Reviewer (if available): _____ **Date:** _____
Signature

PHASE

8**Monitoring & Alerting Configuration**

Days 81-90

Owner: OT SOC Lead

- ☐ **Configure OT SOC dashboard (Grafana/Splunk or equivalent)**
 - Deploy OT-specific dashboard with protocol health, alert counts, compliance status
 - Configure role-based access: operator view, compliance view, executive view
 - Integrate with existing SIEM (Splunk ES, IBM QRadar, or Microsoft Sentinel)

- ☐ **Configure P1 alert: Nation-state TTP or PLC command injection**
 - Trigger: Any command injection attempt, unauthorized unsigned PLC command
 - Response: Immediate notification to Operations Manager and CISO
 - SLA: Notification within 60 seconds of detection

CRITICAL

- ☐ **Configure P2 alert: Unauthorized device on OT network**
 - Trigger: New device not in approved asset inventory communicates on OT network
 - Response: Block device (if non-SIS) and notify OT SOC within 5 minutes
 - SLA: 15-minute response requirement

IMPORTANT

- ☐ **Configure P3 alert: NERC CIP deviation detected**
 - Trigger: Any deviation from documented CIP baseline (ports, configs, access)
 - Response: Notify NERC CIP Compliance Lead and document in incident record
 - SLA: 4-hour remediation or documented exception

- ☐ **Enable physics-aware anomaly alerts**
 - Configure threshold alerts for sensor value violations of physics models
 - Tune thresholds based on 72-hour baseline data (minimize false positives)
 - Verify SIS anomalies escalate to Safety Officer only (no automated block)

IMPORTANT

- ☐ **Configure NIS2 incident reporting automation (EU operators)**
 - Configure NIS2 24-hour early warning report template
 - Configure NIS2 72-hour incident report template
 - Test report generation with simulated incident scenario

- ☐ **Conduct 48-hour monitoring validation with OT SOC team**

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

OT SOC Lead: _____ Date: _____
Signature

Operations Manager: _____ Date: _____
Signature

CISO: _____ Date: _____
Signature

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Compliance Validation

Days 91-105

Owner: NERC CIP Compliance Lead

☐ **Generate complete NERC CIP evidence package (all applicable CIP standards)**

CRITICAL

- CIP-002: BES Cyber System inventory and classification
- CIP-005: ESP monitoring evidence and boundary documentation
- CIP-007: System security management evidence (ports, patches, access)
- CIP-010: Change management baseline and deviation log
- CIP-011: Information protection and encryption evidence

☐ **Conduct IEC 62443 SL-A (achieved security level) assessment**

- Assess each zone against its target Security Level (SL-T)
- Document gap between SL-A and SL-T with remediation plan
- Update zone/conduit documentation with assessed security levels

☐ **Generate NIS2 incident reporting test (EU operators)**

- Simulate a qualifying incident and generate 24h and 72h reports
- Review report content against NIS2 requirements
- Confirm report submission pathway is operational

☐ **Validate audit trail completeness (no gaps in evidence record)**

IMPORTANT

- Verify log retention covers required period (NERC CIP: 6 years)
- Confirm no gaps in continuous monitoring evidence
- Validate timestamping and chain of custody for all evidence

☐ **Conduct TSA Pipeline Security compliance review (pipeline operators)**

- Map QBITEL capabilities to TSA cybersecurity directives
- Generate TSA-formatted compliance evidence package
- Validate incident reporting pathway to TSA

☐ **External reviewer sign-off on evidence quality**

CRITICAL

- Engage qualified NERC CIP subject matter expert for evidence review
- Address any reviewer findings before final sign-off
- Document reviewer credentials and review methodology

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE

NERC CIP Compliance Lead: _____ Date: _____
Signature

External NERC CIP Reviewer: _____ Date: _____
Signature

CISO: _____ Date: _____
Signature

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Operational Handover

Days 106-120

Owner: OT SOC Lead / Operations Manager

- ☐ OT operator training: monitoring, alerting, and incident response (8 hours)
- Alert triage: P1, P2, P3 response procedures
 - SOC dashboard operation and report generation
 - Incident response workflow: contain, investigate, report

IMPORTANT

- ☐ ISSO/CISO training: compliance reporting and evidence management
- NERC CIP evidence package generation and review
 - IEC 62443 assessment update procedure
 - NIS2 and TSA incident reporting workflows

- ☐ Establish 30-day hypercare period with QBITEL engineering support
- Daily QBITEL engineer available during hypercare
 - Define escalation path for any operational issues
 - Document any tuning or configuration adjustments during hypercare

- ☐ Define steady-state NOC/SOC handover procedure
- Transition from hypercare to steady-state support tier
 - Define on-call rotation and escalation contacts
 - Establish change management procedure for ongoing QBITEL updates

- ☐ Document escalation procedures: engineering, safety officer, NERC CIP
- Engineering escalation: equipment issues, timing violations
 - Safety Officer escalation: any SIS anomaly or boundary change
 - NERC CIP escalation: compliance deviation or audit notice

- ☐ Validate complete deployment against Phase 0 scope
- Confirm all intended OT segments are protected
 - Confirm NERC CIP evidence is complete and current
 - Confirm physics model baseline is established for all protected segments

- ☐ **FINAL GO-LIVE SIGN-OFF: All stakeholders required**

CRITICAL

PHASE SIGN-OFF — REQUIRED BEFORE PROCEEDING TO NEXT PHASE	
Operations Manager: _____ Signature	Date: _____
CISO: _____ Signature	Date: _____
Safety Officer: _____ Signature	Date: _____
NERC CIP Compliance Lead: _____ Signature	Date: _____
Asset Owner: _____ Signature	Date: _____

APPENDIX A: Performance SLAs

Metric	SLA Value	Measurement Method	Remediation SLA
PQC overhead	<1ms per transaction	Network tap latency measurement	4 hours
Timing jitter	<100us	Precision timing measurement	4 hours
System availability	99.999% (5 nines)	Uptime monitoring	Immediate (bypass activates)
False positive rate	<0.001%	Alert review and classification	24 hours
Alert delivery: P1	<60 seconds	Alert generation timestamp	Immediate investigation
Alert delivery: P2	<5 minutes	Alert generation timestamp	15-minute response
Alert delivery: P3	<30 minutes	Alert generation timestamp	4-hour remediation
NERC CIP report	<10 minutes	Report generation timestamp	1 hour
Discovery (new tap)	<4 hours	Discovery completion timestamp	24 hours
Threat intel update	Weekly (air-gap transfer)	Update receipt verification	48 hours

APPENDIX B: Rollback Procedures

QBITEL deployment is designed to be fully reversible at each phase. The following emergency procedures enable rapid rollback without operational impact.



Emergency Passthrough Mode: Bypass QBITEL authentication in <60 seconds

CRITICAL

- Access QBITEL management console > Emergency > Passthrough Mode
- Enter dual-authorization codes (two operators required)
- All traffic passes unimpeded - QBITEL continues passive monitoring only
- SOC alert generated and CISO notified automatically
- Passthrough mode logged with timestamp and operator credentials



Phase 5/6 Rollback: Disable PLC command authentication per zone

IMPORTANT

- Access QBITEL policy engine > Zone Configuration
- Set target zone to Monitoring Only mode
- Authentication bypass takes effect immediately (no traffic interruption)
- SCADA polling resumes in unauthenticated mode - same as pre-deployment state
- Document rollback event in change management system



Safety Override Procedure: Emergency SIS communication path

CRITICAL

- SIS communication paths are NEVER blocked by QBITEL under any circumstances
- If SIS communication appears affected, immediately activate Passthrough Mode
- Notify Safety Officer immediately
- Do not attempt to diagnose QBITEL configuration while SIS event is in progress
- QBITEL support: 24/7 emergency line available during hypercare and post-deployment

- ☐ Complete QBITEL Removal: If deployment must be fully reversed
 - Activate Emergency Passthrough Mode
 - Remove QBITEL appliances from network path (tap remains for monitoring)
 - Network returns to pre-deployment state immediately
 - No QBITEL configuration changes affect OT devices - removal is clean
 - Notify QBITEL account team for support and next steps
-

For 24/7 emergency support during deployment, contact: enterprise@qbitel.com | bridge.qbitel.com